Example 2.21

Evaluate $\int \sin^2 x\,dx$

Solution:

Change into simple integrands

$\cos 2x = 1 - 2\sin^2 x$

$\therefore \sin^2 x = \frac{1}{2}(1 - \cos 2x)$

$$\int \sin^2 x\,dx = \int \frac{1}{2}(1 - \cos 2x)\,dx$$

$$= \frac{1}{2}\left[\int dx - \int \cos 2x\,dx\right]$$

$$= \frac{1}{2}\left[x - \frac{\sin 2x}{2}\right] + c$$

Example 2.22

Evaluate $\int \frac{\cos 2x}{\sin^2 x \cos^2 x}\,dx$

Solution:

Change into simple integrands

$$\frac{\cos 2x}{\sin^2 x \cos^2 x} = \frac{\cos^2 x - \sin^2 x}{\sin^2 x \cos^2 x}$$

$$= \frac{1}{\sin^2 x} - \frac{1}{\cos^2 x}$$

$$= \operatorname{cosec}^2 x - \sec^2 x$$

$$\int \frac{\cos 2x}{\sin^2 x \cos^2 x}\,dx = \int \left(\operatorname{cosec}^2 x - \sec^2 x\right)dx$$

$$= -\cot x - \tan x + c$$

Example 2.23

Evaluate $\int \sqrt{1 + \sin 2x}\,dx$

Solution:

Change into simple integrands

$1 + \sin 2x = \sin^2 x + \cos^2 x + 2\sin x \cos x$

$= (\sin x + \cos x)^2$

$$\int \sqrt{1 + \sin 2x}\,dx = \int \sqrt{(\sin x + \cos x)^2}\,dx$$

$$= \int (\sin x + \cos x)\,dx$$

$$= -\cos x + \sin x + c$$

Example 2.24

Evaluate $\int \cos^3 x\,dx$

Solution:

Change into simple integrands

$\cos 3x = 4\cos^3 x - 3\cos x$

$\cos^3 x = \frac{1}{4}[\cos 3x + 3\cos x]$

$= \frac{1}{4}\cos 3x + \frac{3}{4}\cos x$

$$\int \cos^3 x\,dx$$

$$= \frac{1}{4}\int \cos 3x\,dx + \frac{3}{4}\int \cos x\,dx$$

$$= \frac{\sin 3x}{12} + \frac{3\sin x}{4} + c$$

Exercise 2.4

Integrate the following with respect to x.

1. $2\cos x - 3\sin x + 4\sec^2 x - 5\operatorname{cosec}^2 x$
2. $\sin^3 x$
3. $\frac{\cos 2x + 2\sin^2 x}{\cos^2 x}$
4. $\frac{1}{\sin^2 x \cos^2 x}$ $\left[\text{Hint}: \sin^2 x + \cos^2 x = 1\right]$
5. $\sqrt{1 - \sin 2x}$

2.1.4 Integration by parts

Type: V

(1) We know that, if u and v are two differentiable functions of x, then

$$\frac{d}{dx}(uv) = u\frac{dv}{dx} + v\frac{du}{dx}.$$

Integrate on both sides with respect to x.

$$\int \frac{d}{dx}(uv)\,dx = \int u\frac{dv}{dx}\,dx + \int v\frac{du}{dx}\,dx$$

$$\Rightarrow \int d(uv) = \int u\,dv + \int v\,du$$

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